NTNU

Department of Industrial Economics and Technology Management

Examination paper for TIØ4145 Corporate Finance

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Other information:

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Informasjon om trykking av eksamensoppgave

Originalen er:

1-sidig ☐ **2-sidig** ☐

sort/hvit ☐ **farger** ☐

skal ha flervalgskjema ☐

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Date Signature

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Task 1 - Stocks and Bonds [20%]

a. Suppose you are trying to determine the interest rate sensitivity of two bonds. Bond 1 is a 12% coupon bond with yearly payouts, a 7-year maturity and a $1000 face value. Bond 2 is a 'zero-coupon' bond that pays $1000 after 7 year. The current interest rate is 12%.
   i. Determine the duration of each bond.
   ii. Which bond is more sensitive to interest rate changes? Explain.

	Cash Flow		PV(r=12%)	
Year	Bond 1	Bond 2	Bond 1	Bond 2
1	120	0	107.14	-
2	120	0	95.66	-
3	120	0	85.41	-
4	120	0	76.26	-
5	120	0	68.09	-
6	120	0	60.80	-
7	120	0	54.28	-
7	1000	1000	452.34	452.34
P_0			1000.00	452.34

Duration bond 1:

$$\frac{[107.14 + 95.66 \cdot 2 + 85.41 \cdot 3 + 76.26 \cdot 4 + 68.09 \cdot 5 + 60.80 \cdot 6 + 507.63 \cdot 7]}{1000} = 5.11139$$

Duration bond 2:

$$\frac{452.34 \cdot 7}{452.34} = 7$$

Bond 2 will be more sensitive to interest rate changes.

b. The total value of a firm is 100 ($ million). The dividend policy of the firm is a 5% dividend yield; its debt policy is to be 75% equity financed. Investors expect a 16% return on stocks of firm A, and ask 4% interest on firm A's debt.
   i. What constant expected growth rate of the firm is in line with the numbers (assuming no new equity will be issued)?
   ii. What then are the expected earnings in year 1 and in year 5?

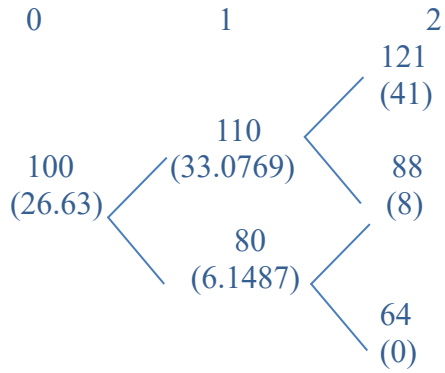
From data for equity we have: $g = r - DIV_1 / P_0 = 16\%-5\% = 11\%$. This must also be the growth rate of the firm if the D/E ratio is kept fixed (and no new equity is issued) Expected earnings in year 1 are 13 (as $r_A = 0.25 \times 4\% + 0.75 \times 16\% = 13\%$); alternatively: for interest: $0.25 \times 100 \times 4\% = 1$, for dividends $0.75 \times 100 \times 5\% = 3.75$, for growth $0.75 \times 100 \times 11\% = 8.25$ from retained earnings (the rest of growth is by extending debt with $0.25 \times 100 \times 11\% = 2.75$). In year 5: expected growthrate of everything is 11% annually, so expected earnings year 5 $= 1.11^4 \times$ earnings year 1 = 19.735.

Task 2 - Financial Options [30%]

Consider a two-year European call option on Pineapple Inc. stock. The current stock price is 100 USD. In each of the next two years the stock price will either increase by 10% or decrease by 20%. The strike price of the call is 80 USD. The stock pays no dividends and the risk free interest rate is equal to 4%.

a. Using a two-period binomial model, calculate the price of this European call option.

Binomial tree



b. Calculate the price of the same European call option using risk-neutral valuation. Show the steps explicitly.

Risk neutral probability of an up state is equal to $\frac{1.04*100-80}{110-80} = 0.8$

Call option price is $\frac{0.8*0.8*41+2*0.2*0.8*8}{1.04*1.04} = 26.63$

c. Compare the outcomes in *a.* and *b.* and comment on the result.

They are the same (differences may arise due to rounding up error). Binomial model does not require assumptions on investor's preferences to risk, so it should also work for risk-neutral preferences.

Suppose that the Stock Exchange also listed an American call option with the same characteristics.

d. Did such a decision make sense if Pineapple Inc. was not expected to pay out dividends or repurchase shares in the foreseeable future? Explain your answer.

No. It is never optimal to exercise early an American option on a non-dividend-paying stock.

e. What is the price for this American option, and how does it compare to the price of the European option? Explain.

The same as the price of European as there is no value in early exercise.

f. How would your answer in e. change if Pineapple Inc. decided to pay out dividends? Explain intuitively.

Depends if it is valuable to exercise the option yearly, if there is value than American will be worth more than European.

Task 3 - Capital Structure [10%]

As a financial consultant, you are given the following information about two companies, one levered and one unlevered:

	Unlevered	Levered
Before tax operating income	$10 000	$10 000
Interest on debt	0	2500
Cost of capital (WACC)		16%
Cost of Equity capital		
Cost of Debt capital		
Value of equity		22 500
Value of debt		
Value of fim		

Copy the table to your answer sheet and fill in the missing figures. Show the steps you used to calculate them. Assume that the corporate tax rate is 40% and all cash flows are perpetual.

	Firm U	Firm L
Before tax operating income	10,000	10,000
Interest on debt	0	2,500
Cost of capital	19.05%	16.0%
Cost of Equity capital	19.05%	20.0%
Cost of Debt capital	-	16.67%
Value of equity	31,500	22,500
Value of debt	0	15,000
Value of firm	31,500	37,500

$$V_L = \frac{\$10,000(1-0.4)}{0.16} = \$37,500$$
$$D_L = V_L - E_L = \$15,000$$
$$D_L = \$15,000 = \frac{\$2,500}{r_D}$$
$$\Rightarrow \quad r_D = 16.67\%$$
$$E_L = \$22,500 = \frac{(10,000-2,500)(1-0.40)}{r_E}$$
$$\Rightarrow \quad r_E = 20\%$$
$$r = 16\% = 0.20\frac{22,500}{37,500} + 0.1667(1-0.4)\frac{15,000}{37,500}$$
$$V_U = V_L - \tau^c D = \$37,500 - 0.4 \cdot \$15,000 = \$31,500$$
$$V_U = E_U = 31,500 = \frac{10,000(1-0.4)}{r_E}$$
$$\Rightarrow \quad r_E = r = 19.05\%$$

Task 4- Portfolio Choice [25%]

Suppose that there are two stocks available in the market: stock A and stock B. The expected returns of these stocks are equal, and given by $E[R_A]=E[R_B]=0.16$. The standard deviation of stock A is equal to 0.4, and the standard deviation of stock B is 0.7. The stocks are positively correlated with the correlation coefficient equal to 0.18.

a. Find the expected return and volatility of the portfolio consisting of 20% of stock A and 80% of stock B.
Expected return 0.16 and variance is $(1-x)^2 s1^2 + x^2 s2^2 + 2x(1-x)\rho s1s = 0.3424$. Volatility is 0.59

b. Is this portfolio efficient? Explain.
No, you can achieve the same expected return and a smaller variance for example by just investing everything in stock A.

c. Sketch the portfolios that one can create by combining stocks A and B for the following weights:

Weight Stock A	Weight Stock B
0%	100%
20%	80%
60%	40%
80%	20%
100%	0%

Illustrate the efficient frontier.
The portfolios will end up on the line, and the efficient frontier will be one point in this case.

d. What is the optimal portfolio choice for a typical risk-averse investor?
Risk averse investor will choose a minimum variance portfolio investing approx. 17,6% in stock B and 82,4% in stock A

e. Explain intuitively what correlation coefficient represents.
The degree of co-movements in stock returns

f. How would your answer to question *d.* change if the correlation coefficient changed? Consider two extreme examples: when stocks A and B are perfectly positively correlated and when they are perfectly negatively correlated.

If perfectly positively, you invest everything is stock A because it has a smaller standard deviation. Is perfectly negatively the new optimal portfolio weights are 36.4% stock B and 63.6% in stock A, so you add more of stock B to benefit from diversification.

Task 5 - Pricing Theories [15%]

In the discussions of some of the models in this course, we relied on the following type of argument: "If two investment strategies have the same payoff tomorrow, they must have the same price today."

a. What did we call this statement in the course? State the assumptions that are necessary for this statement to hold.
Law of one price, no arbitrage argument

b. For which of the following theories, shortly explaining each theory and explain which of the following theories relies explicitly on no-arbitrage arguments.
   i. The Put Call parity of option pricing.
   ii. The Miller Modigliani propositions for capital structure.
   iii. The CAPM.

For i and ii.

i. The equation for Put Call parity comes from the fact that the combinations of stock with a put and bond with a call have the same payoffs, thus, they must have to have the same price.
ii. In the absence of taxes or other transaction costs, the total cashflow to all of the firm's security holders is equal to the total cashflow generated by the firm's assets. Thus, by law of one price securities and assets must have the same market value. (p.525 in the book)
iii. CAMP does not explicitly rely on no arbitrage argument. It is an equilibrium theory, where it is required that all investors hold market portfolio.